draftsmith — single-shot vs phased loop

both conditions use the design-guidelines prompt · 2026-07-30

Paired within model per brief. Loop = plan -> perimeter -> rooms one-per-turn -> refine, phase-gated findings, ?-queries; capped at 25 model calls in this experiment.

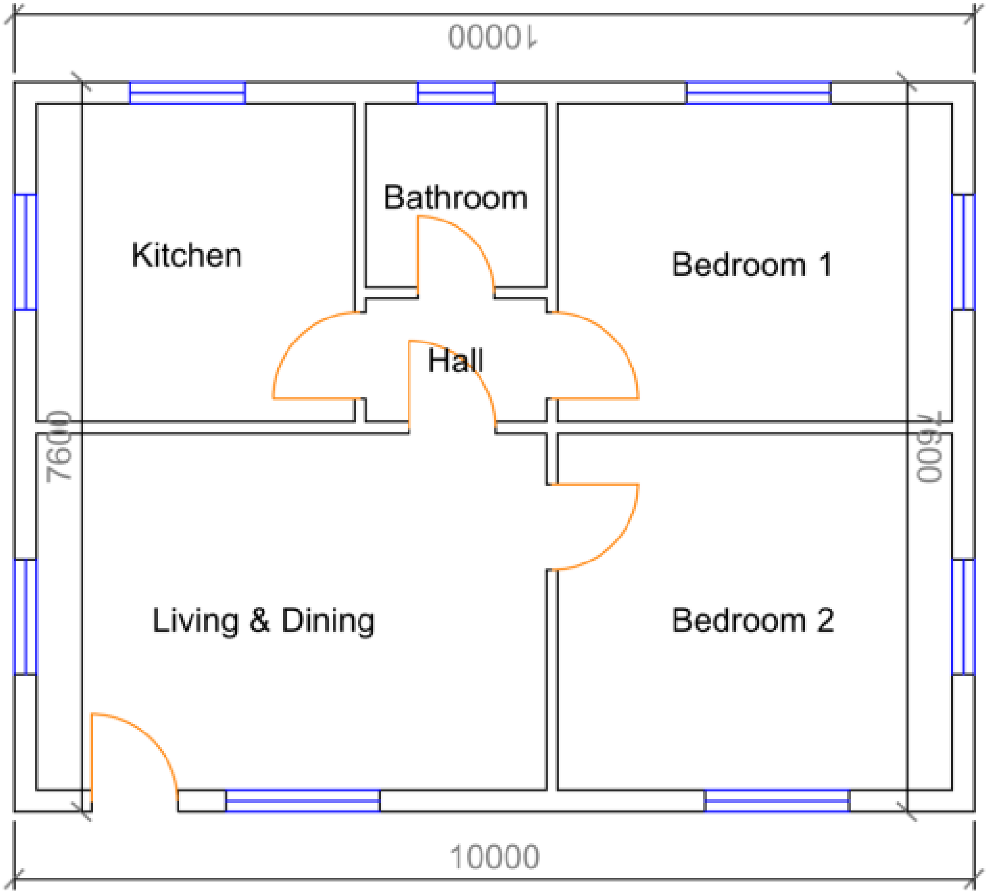
brief	model	cond	C	B	S	calls	hard
2bhk-terse	3.6-flash	single	1.00	1.00	0.86	1	-
2bhk-terse	3.6-flash	loop	1.00	1.00	0.91	9	-
3bhk-dimensioned	3.1-flash-lite	single	0.73	0.25	0.74	1	door_economy
3bhk-dimensioned	3.1-flash-lite	loop	0.59	0.57	0.80	25	reachability,habitable_ventilation
4bhk-passage	3.1-flash-lite	single	0.62	0.14	0.77	1	door_economy,bedroom_privacy
4bhk-passage	3.1-flash-lite	loop	0.38	0.71	0.63	25	reachability,entrance,door_economy,habi

mean single C 0.784 B 0.464 S 0.788
mean loop   C 0.658 B 0.761 S 0.781

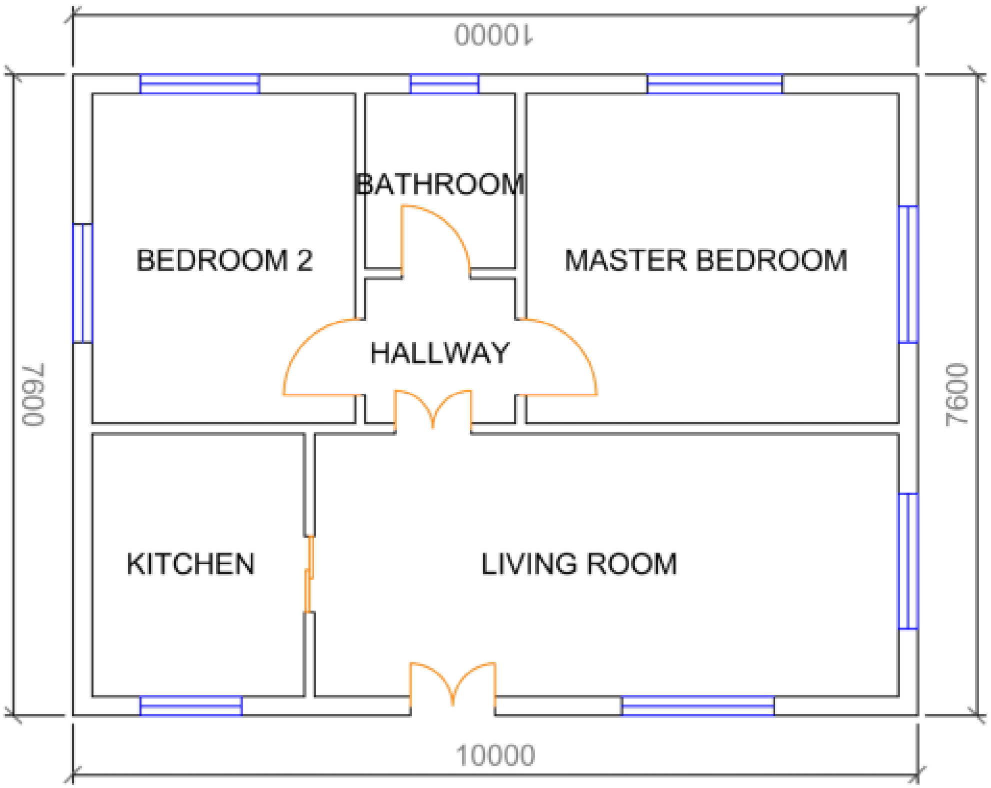
Reading:
- Compliance (brief adherence) is the loop's big win: the plan phase forces a complete program before geometry (+0.30 mean; 4bhk 0.14 -> 0.71 on the weak model).
- Loop correctness losses are truncation artifacts: both lite loop runs hit the 25-call cap mid-session, leaving rooms unconnected. On the capable model (2bhk pair) the completed loop beats single-shot everywhere it differs.
- The plan gate caught real under-planning (bath count) and now waives with a recorded violation instead of aborting.
- Next: completion-aware budget (reserve calls to close the plan), and rerun on the full-strength model when quota allows.

2bhk-terse · gemini-3.6-flash

single-shot (1 calls)
C 1.00 · B 1.00 · S 0.86 · hard: none

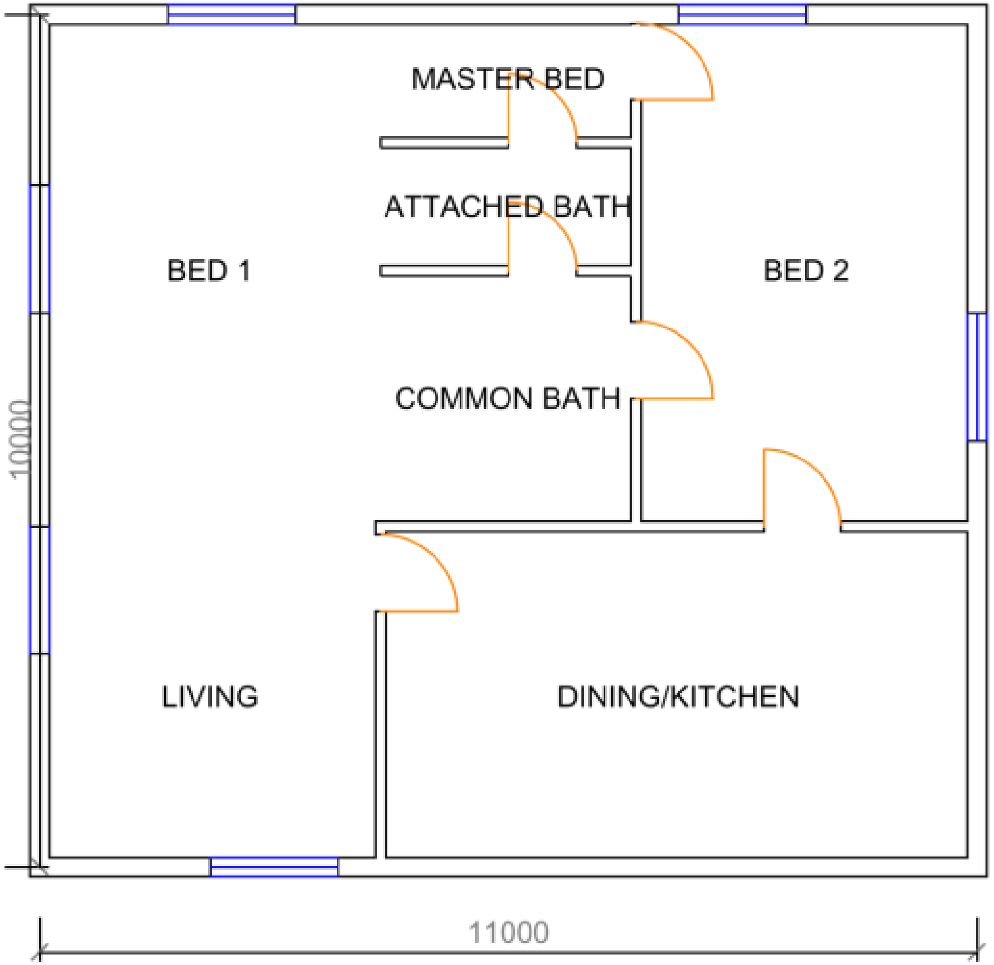


loop (9 calls)
C 1.00 · B 1.00 · S 0.91 · hard: none

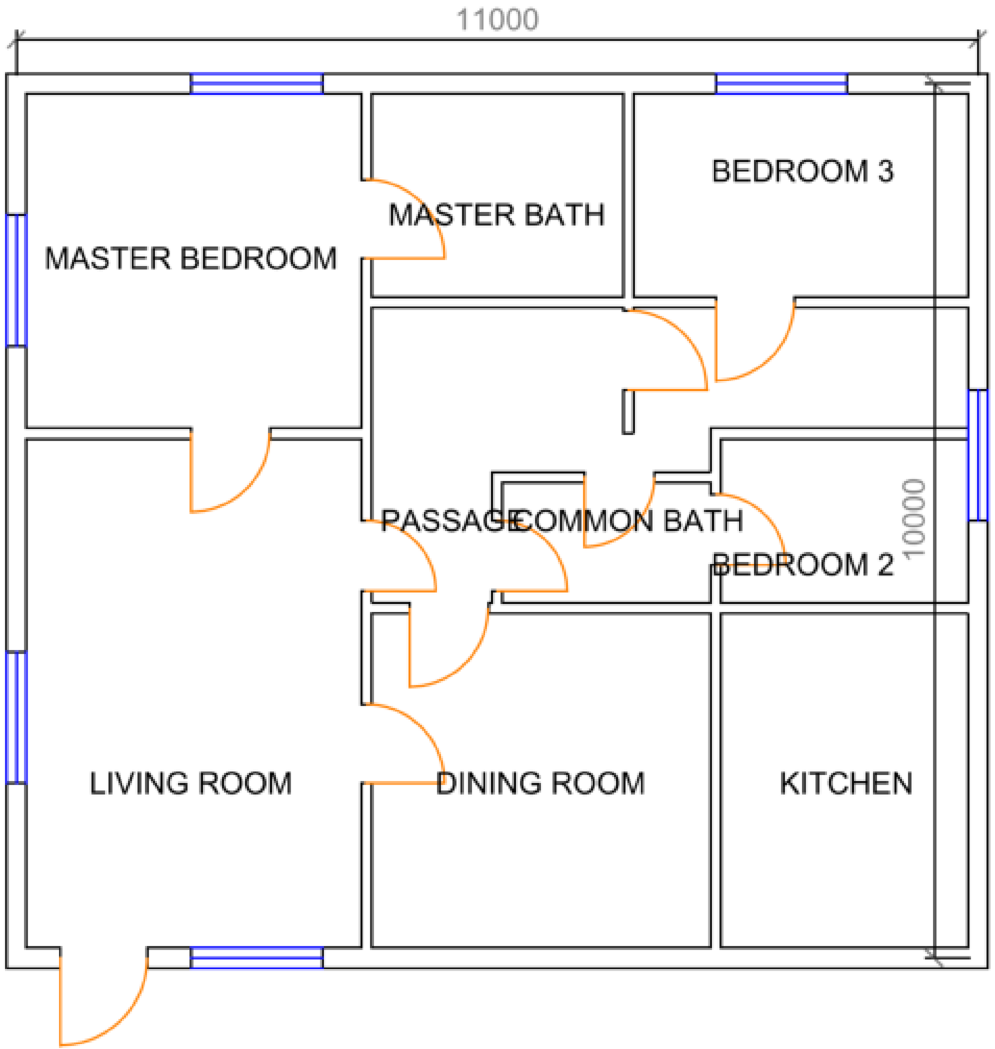


3bhk-dimensioned · gemini-3.1-flash-lite

single-shot (1 calls)
C 0.73 · B 0.25 · S 0.74 · hard: door_economy

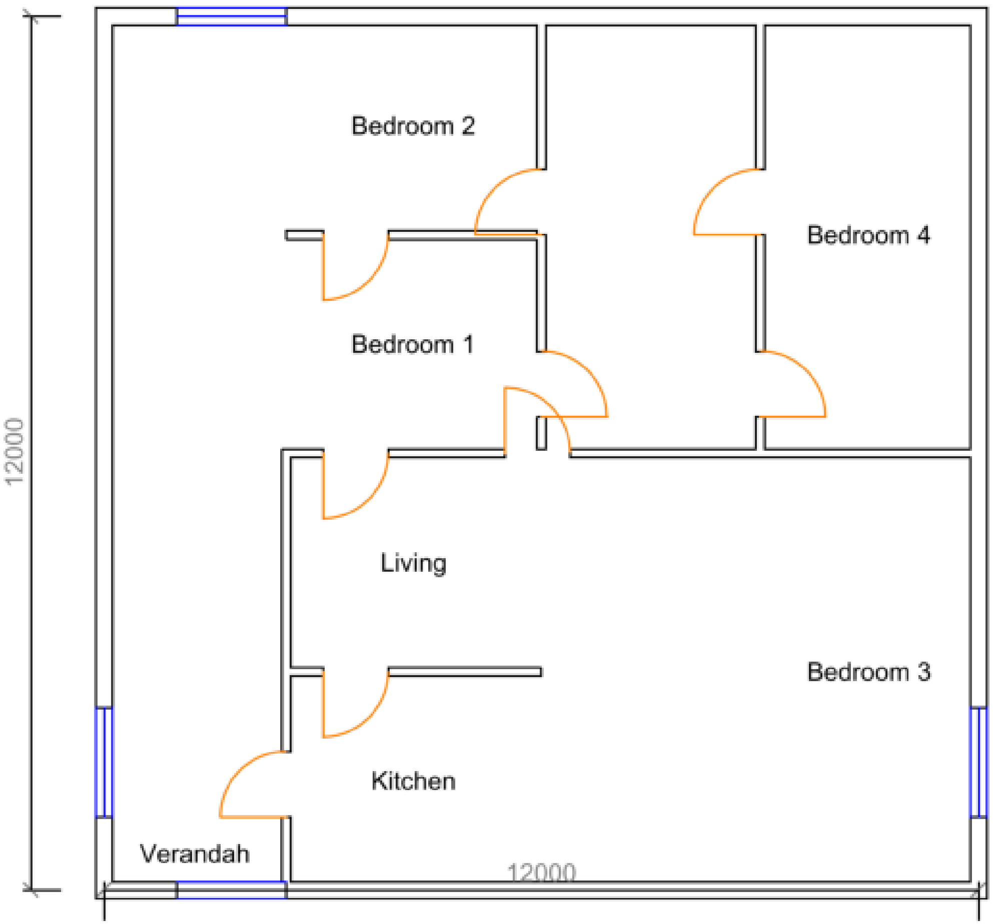


loop (25 calls)
C 0.59 · B 0.57 · S 0.80 · hard: reachability, habitable_ventilation



4bhk-passage · gemini-3.1-flash-lite

single-shot (1 calls)
C 0.62 · B 0.14 · S 0.77 · hard: door_economy, bedroom_privacy



loop (25 calls)
C 0.38 · B 0.71 · S 0.63 · hard: reachability, entrance, door_economy, habitable_ventilat